HAFIZ HILAL AHMAD

MS Structural Engineering · Civil Engineer (PEC Reg. No. Civil/59341)

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RESEARCH STATEMENT

Structural engineer with hands-on experimental research experience in the seismic behaviour of confined masonry and an industry background in large-scale infrastructure construction. Holds an MS in Structural Engineering from NUST Islamabad, with thesis work on the dynamic performance of a confined asymmetrical masonry building investigated through full-scale shake-table testing and calibrated finite-element analysis in ABAQUS. Research interests center on large-scale experimental testing of masonry structures, the analysis and earthquake-resistant design of masonry construction, numerical modelling of nonlinear masonry behaviour, and the development of sustainable, low-carbon masonry systems for seismic regions in low- and middle-income countries.

EDUCATION

MS in Structural Engineering · National University of Sciences and Technology (NUST), Islamabad

October 2023 – September 2026 · CGPA: 3.50 / 4.00

- MS thesis: "Dynamic Performance of a Confined Asymmetrical Masonry Building — Shake-Table Testing and Finite-Element Analysis."
- Relevant coursework: Advanced Structural Dynamics, Earthquake Engineering, Finite Element Analysis, Advanced Concrete Structures, Structural Stability.

BS in Civil Engineering · International Islamic University, Islamabad (IIUI)

2016 – 2020 · CGPA: 3.07 / 4.00

- Final-year project: Structural fire rating of an office building in Islamabad (structural fire engineering).
- Foundational coursework in Structural Analysis, Reinforced Concrete Design, Steel Design, Geotechnical Engineering, and Construction Management.

Higher Secondary School Certificate (FSc – Pre-Engineering) · HITEC College, Taxila Cantt.

Completed 2015 · Grade: A

Secondary School Certificate (SSC – Science) · HITEC College, Taxila Cantt.

2010 – 2012 · Grade: A1

RESEARCH EXPERIENCE

Graduate Researcher · Department of Structural Engineering, NUST Islamabad

Oct 2023 – Present

- Investigating the dynamic seismic performance of a confined asymmetrical masonry building through scaled shake-table testing, with the aim of characterizing its global response, torsional behaviour, damage progression, and failure mechanisms under recorded ground motions.
- Developing parallel finite-element (FEM) models calibrated against the shake-table results, in order to validate numerical predictions, quantify modelling uncertainties, and extend the experimental findings through parametric studies.
- Hands-on experience with experimental setup design, instrumentation (accelerometers, LVDTs, strain gauges), data acquisition, signal processing, and post-test damage characterization.
- Developing competence in nonlinear time-history analysis and macro/micro-modelling of masonry in ABAQUS, with a focus on confined-masonry constitutive behaviour and asymmetric in-plan response.

Undergraduate Research Assistant – Structural Fire Engineering · IIU Islamabad

2019 – 2020

- Investigated structural fire rating of an office building, including assessment of fire load and probabilistic temperature exposure of structural members.

PUBLICATIONS

1. M. Noman, **H. H. Ahmad**, A. Hamza, and M. Yaqub, "Assessment of Fire Load and Probabilistic Temperature for Office Buildings in Pakistan," *Civil Engineering Infrastructures Journal*, pp. 1–27, 2022. DOI: 10.22059/CEIJ.2022.341199.1827.

RESEARCH INTERESTS

- Large-scale experimental testing of masonry and structural systems, including shake-table and quasi-static testing.
- Analysis and earthquake-resistant design of masonry structures (confined, unreinforced, and retrofitted masonry).
- Nonlinear finite-element modelling of masonry under seismic loading, including torsional and in-plan asymmetric response.
- Sustainable and low-carbon masonry construction (including earth masonry and other low-embodied-energy systems) for seismic regions.

PROFESSIONAL EXPERIENCE

Laboratory Engineer · Military College of Engineering MCE (Structural Dynamics Lab) Risalpur, Pakistan

November 2025 – Present

Assistant Project Manager · Frontier Works Organization (FWO), Pakistan

April 2022 – October 2023 (1 year, 7 months)

- **Rawalpindi Ring Road Project (RRR), Phase-I** — 38.3 km from Banth (N-5) to Thalian Interchange (M-2). Supervised earthwork, structural works, and on-site planning under the 751 CTE FWO command.
- **Kurram Tangi Multipurpose Dam Project, Phase-I** (764 CTE FWO) — supervised construction of tunnels, canals, and water-retaining structures for an 18.4 MW hydropower and 16,300-acre irrigation scheme.
- Coordinated multidisciplinary site teams, monitored compliance with construction standards, and contributed to project planning and progress reporting.

Site Engineer · Alzeb Burki Construction Company (SMC – Pvt) Ltd

June 2021 – March 2022 (10 months)

- Sub-contractor to FWO (763 CTE) on the Thall–Mirali Road (54 km) project in North Waziristan.
- Ensured timely execution of scheduled activities, quality control on site, and adherence to engineering specifications. Supported project-management functions including progress monitoring and reporting.

Construction Inspector (Trainee) · EA Consulting (Pvt) Ltd

July 2019 – August 2019 (2 months)

- Site inspector during construction of a high-rise residential development in G-13, Islamabad.

TECHNICAL SKILLS

Structural analysis & FEM: CSI ETABS, CSI SAFE, Abaqus, SAP2000 (basic).

BIM / CAD: AutoCAD, Autodesk Revit.

Programming & data: Python, Microsoft Excel (advanced).

Productivity: Microsoft Word, PowerPoint; LaTeX (basic).

Languages: English (professional working proficiency), Urdu (native), Pashto (native).

LICENCES, CERTIFICATIONS & MEMBERSHIPS

- Registered Engineer — Pakistan Engineering Council (PEC), Reg. No. Civil/59341.
- Former member, Islamic Engineering Society, IIU Islamabad.

ADDITIONAL ACTIVITIES

- Selected participant, Cultural and Educational Exchange Program to the North Carolina United States (HITEC College, 2013).

REFEREES

Available upon request.